

## 1. What is Seaborn? How is it different from Matplotlib?

What is Seaborn?

Seaborn is a Python data visualization library built on top of Matplotlib. It is mainly used for statistical data visualization. It provides high-level functions to create attractive and informative statistical graphs with very less code.

Seaborn works very well with Pandas DataFrames and helps in visualizing relationships, distributions, and categorical comparisons easily.

Difference between Seaborn and Matplotlib:

Matplotlib:

- Low-level plotting library.
- Requires more code for styling and customization.
- Provides full control over plot elements.
- Suitable for basic and custom plots.

Seaborn:

- Built on top of Matplotlib.
- Provides better default styles and themes.
- Easier syntax for complex statistical plots.
- Directly works with Pandas DataFrames.
- Ideal for statistical visualization.

In simple words:

Matplotlib gives you full control.

Seaborn makes statistical visualization easier and more attractive.

## 2. How to install Seaborn library?

Using pip:

```
pip install seaborn
```

Using conda:

```
conda install seaborn
```

To check installation:

```
import seaborn as sns  
print(sns.version)
```

### 3. Difference between countplot() and barplot()

countplot():

- Used to count the number of occurrences of each category.
- Works automatically on categorical data.
- Does not require y-values.
- It shows frequency.

Example:

```
sns.countplot(x="gender", data=df)
```

barplot():

- Used to show the average (mean) of a numerical variable for each category.
- Requires both x (category) and y (numeric value).
- Performs aggregation (mean by default).

Example:

```
sns.barplot(x="gender", y="salary", data=df)
```

Main difference:

countplot shows count.

barplot shows aggregated value (mean by default).

### 4. How to create a boxplot in Seaborn?

A boxplot shows distribution, median, quartiles, and outliers.

Example:

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
sns.boxplot(x="department", y="salary", data=df)
```

```
plt.show()
```

Boxplot shows:

- Minimum
- Q1 (25%)
- Median (50%)
- Q3 (75%)
- Maximum
- Outliers

### 5. What is the use of hue parameter?

The hue parameter is used to divide data into subgroups using different colors.

It adds another categorical dimension to the plot.

Example:

```
sns.barplot(x="department", y="salary", hue="gender", data=df)
```

Here:

- Bars are grouped by department.
- Colors represent gender.

Hue helps compare categories within categories.

### 6. What is kdeplot?

kdeplot stands for Kernel Density Estimation plot.

It shows the probability density of a continuous variable.

It is like a smooth version of a histogram.

Example:

```
sns.kdeplot(data=df, x="salary", fill=True)
```

It helps understand:

- Distribution shape
- Skewness
- Peaks

### 7. What is the use of jointplot?

jointplot shows the relationship between two numerical variables.

It combines:

- Scatterplot (center)
- Histogram or KDE (margins)

Example:

```
sns.jointplot(x="age", y="salary", data=df, kind="scatter")
```

It helps analyze:

- Correlation
- Distribution

- Trend between two variables

#### 8. What is pairplot()? What is the importance of pairplot?

pairplot creates scatter plots for all pairwise combinations of numerical variables in a dataset.

Example:

```
sns.pairplot(df)
```

It automatically:

- Plots scatterplots for every pair of features
- Shows distribution on diagonals

Importance:

- Helps in exploratory data analysis (EDA)
- Detects relationships
- Detects outliers
- Identifies patterns
- Useful before building ML models

#### 9. Difference between distplot() and histplot()

distplot():

- Older function.
- Used to combine histogram and KDE.
- Now deprecated in newer versions.

histplot():

- New function.
- More flexible and powerful.
- Can show histogram, KDE, or both.
- Recommended instead of distplot.

Example:

```
sns.histplot(df["salary"], kde=True)
```

Main difference:

distplot is outdated.

histplot is the modern replacement.

## 10. How to create heatmap?

Heatmap is used to visualize matrix data using colors.

Example:

```
import seaborn as sns
import matplotlib.pyplot as plt

data = df.corr()
sns.heatmap(data, annot=True, cmap="coolwarm")
plt.show()
```

Parameters:

- `annot=True` → shows values inside boxes
- `cmap` → changes color style

## 11. Different categories available in Seaborn

Seaborn plots are mainly divided into:

### 1. Relational plots

- `scatterplot()`
- `lineplot()`

### 2. Categorical plots

- `barplot()`
- `countplot()`
- `boxplot()`
- `violinplot()`
- `stripplot()`
- `swarmplot()`

### 3. Distribution plots

- `histplot()`
- `kdeplot()`
- `ecdfplot()`

### 4. Matrix plots

- `heatmap()`
- `clustermap()`

### 5. Regression plots

- `regplot()`
- `lmpplot()`

## 6. Multi-plot grids

- `pairplot()`
- `jointplot()`
- `FacetGrid()`

## 12. How to create a histplot?

Example:

```
sns.histplot(data=df, x="salary", bins=10, kde=True)
plt.show()
```

Important parameters:

- `bins` → number of intervals
- `kde=True` → adds smooth curve
- `color` → change color
- `multiple` → stack or dodge for categories

## 13. How to visualize correlation matrix?

Step 1: Calculate correlation

```
corr = df.corr()
```

Step 2: Plot heatmap

```
sns.heatmap(corr, annot=True, cmap="coolwarm", linewidths=0.5)
plt.show()
```

This helps to:

- Identify strong positive correlation (close to 1)
- Identify strong negative correlation (close to -1)
- Detect multicollinearity
- Feature selection for ML

Seaborn is a high-level statistical visualization library built on Matplotlib. It simplifies complex visualizations, provides better aesthetics, and integrates smoothly with Pandas. It is widely used in exploratory data analysis, machine learning, and data science projects.